BROOKE MURPHY

San Luis Obispo, California | bmurph32@calpoly.edu

RESEARCH INTERESTS

Groundwater hydrology, remote sensing, hydrologic modeling, GIS and spatial analysis, agricultural water management, drought and water-resource decision support

EDUCATION

Master of Science, Environmental Science and Management — California Polytechnic State University, San Luis Obispo Expected Dec. 2026

- Advisor: Dr. Yiwen Chiu. Graduate GPA: 3.8.
- Agricultural Research Institute NextGen fellowship for "Analyzing the Spatiotemporal Transferability of a Remote Sensing-Informed Machine Learning Model for ETo"
- Selected coursework: Applied GIS, Advanced Environmental Management, Research Methods & Data Analysis, Advanced Applications in GIS, Water Resource Law and Policy, Groundwater Hydraulics & Hydrology Engineering.

Bachelor of Arts, Environmental Studies — Westmont College, Santa Barbara, California May 2023

- Graduated magna cum laude; cumulative GPA 3.8; Honors Program.
- Selected coursework: Environmental Studies of the Natural World; Forest Management for a Sustainable World; Permaculture Design; Environmental Sociology; Environmental Ethics.

Bachelor of Arts, Religious Studies — Westmont College, Santa Barbara, California May 2023

- Selected coursework: Reformation & Modern Christianity; Early & Medieval Christianity; World Religions; Christian Apologetics; Gender in Theological Perspectives; Greek I & II.

RESEARCH EXPERIENCE

Graduate Researcher — Evaluating Remote Sensing-Based Reference Evapotranspiration — Cal Poly, San Luis Obispo 2025–Present

- Compare methods of estimating reference evapotranspiration for agricultural water-use planning using temporal and spatial analysis.
- Integrate climate, remotely sensed, and spatial datasets to evaluate model differences across landscapes.

Groundwater Management Intern — Valley Water, Santa Clara County, California Jun.–Aug. 2026

- Analyzed and quality-checked groundwater datasets, reviewed historical well records, updated databases, and developed Esri StoryMaps on seawater intrusion, sea-level rise, and land subsidence.
- Supported groundwater-quality monitoring through PFAS soil-core logging, recharge-pond sampling, laboratory coordination, and water-sample collection for the Domestic Well Testing Program.

Natural Resources Management Intern — Santa Clara County Parks Jun.–Aug. 2022

- Directed a study on restoration measures for *Asclepias californica* populations on public lands.
- Partnered with CalFire to support prescribed burns in high-risk areas on public lands.

Biological Monitoring Volunteer — Bureau of Land Management, Cascade-Siskiyou National Monument, Oregon Aug.–Dec. 2020

- Conducted research on soil horizons, conifer forest health, and biodiversity in clear-cut forest areas.

TEACHING & TUTORING EXPERIENCE

Instructional Student Assistant — **Introductory Environmental Science** — California Polytechnic State University, San Luis Obispo Fall 2026
Instructor: Dr. Nick Williams

- Support instruction and student learning in an introductory environmental science course.

Teaching Assistant & Tutor — **Old Testament** — Westmont College, Santa Barbara, California Spring 2023
Instructor: Dr. Sandra Richter

- Assisted with an undergraduate Old Testament course and led regular academic tutoring sessions for all students.

PROFESSIONAL EXPERIENCE

GIS Developer — California Polytechnic State University, San Luis Obispo Oct. 2026–Present

- Restructure the information architecture and navigation of an open-source introductory GIS course website to improve readability
- Develop and maintain course content in Markdown using MkDocs Material, with version control, collaboration, and deployment through Git, GitHub, and GitHub Pages.

Program Coordinator — Santa Barbara County Parks, Cachuma Lake Apr. 2024–Jan. 2025

- Conducted plankton tows, substrate sampling, vessel inspection, and vessel decontamination under the California State Parks Aquatic Invasive Species Mitigation Program.
- Organized educational events and coordinated program procedures with resource-management agencies, volunteers, and the public.

TECHNICAL SKILLS

GIS & Geospatial: ArcGIS Pro, ArcGIS Online, StoryMaps, Field Maps, QGIS, Google Earth Engine, ModelBuilder, georeferencing, raster analysis, geodatabase management, cartographic design
Programming & Data: R (primary), Python, SQL, JavaScript (basic), Excel, Git/GitHub
Water & Environmental: Water-footprint accounting, watershed assessment, SGMA framework familiarity, NPDES permit analysis, groundwater data QA/QC, soil-core logging, aquatic invasive species mitigation, plankton tow sampling, biological monitoring.

HONORS & FUNDING

- Wildung Family Scholarship, 2026
- Harold G. Hull Graduate Assistantship, Fall 2026
- Agricultural Research Institute NextGen Fellowship
- Magna cum laude, Westmont College
- Augustinian Scholarship, Westmont College
- Dean’s List at Westmont College